Applied Psychophysiology and Biofeedback: Translating Evidence into Practice

Credit Value: 4 ECTS | Course Format: Blended Learning

Course phases:

Phase 1: Online (asynchronous + team coordination)

Phase 2: On-site 5-day intensive (Reykjavik University)

Phase 3: Online capstone (analysis, report, and final presentations)

Target audience

Master's and PhD students in psychology, neuroscience, biomedical engineering, health sciences, and related disciplines.

Dates, duration and location

Course dates: 21 May - 17 June

Online participation: 21 May - 31 May and 6 June - 17 June (asynchronous modules + scheduled online sessions).

On-site participation: 1 June - 5 June (5-day intensive).

On-site teaching location: Reykjavik University, Reykjavik, Iceland.

Students aiming to receive 4 ECTS are expected to participate in online components, complete assignments, and finish a capstone project (written report + oral presentation).

Hosting institution

Reykjavik University, by the Institute for Biomedical and Neural Engineering.

Program fees and funding

No fee for applicants from NeurotechEU partner institutions.

Applicants outside the NeurotechEU consortium: 450 EUR participation fee (excluding travel and accommodation).

The sending university bears the costs for accommodation and travel.

Contacts

For pre-course administrative issues: Ragnheidur Þórhallsdóttir (nteu@ru.is)

For course-specific questions: Sigrun Þóra Sveinsdottir (sigrunths@ru.is)

How to apply – Deadline X Xth, 2026

For students from NeurotechEU partners, apply through your local NeurotechEU contact person by sending them the completed application form.

For non-NeurotechEU students, contact Ragnheiður Þórhallsdóttir: ragnheidurth@ru.is at Reykjavik University**.**

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1. Course Description

This 4 ECTS blended course provides a comprehensive introduction to the interdisciplinary field of Applied Psychophysiology and Biofeedback. It equips students with both theoretical foundations and practical competencies in measuring, interpreting, and applying psychophysiological signals - primarily Heart Rate Variability (HRV) - to improve mental and physical performance, well-being, and clinical outcomes.

The course begins with a wide-angle lens on applied psychophysiology, introducing students to various tools and approaches, including HRV, neurofeedback, interoception training, and emerging bio-sensing technologies. The focus then narrows, then narrows to HRV biofeedback (HRVB): physiological mechanisms, evidence base, and practical implementation. Students will follow a structured HRV self-regulation training program throughout the course (daily practice supported by an app and an individual workbook).

During the 5-day on-site intensive, students work in interdisciplinary teams to collect psychophysiological data, experience multiple intervention components (e.g., resonance breathing, compassion-based practices, and cognitive demand/stress tasks), and run a small team experiment. The course culminates with a capstone project directly derived from this in-person experimentation phase. Students will analyze their group's intervention data and design an evidence-based mini-intervention, culminating in a written report and an oral presentation that demonstrate their ability to synthesize practical findings with scientific rationale, ethical considerations, and applied design.

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2. Learning Outcomes

Upon successful completion of the course, students will be able to:

- Describe the scope and methods of applied psychophysiology.

- Explain physiological principles of HRV and its role in emotion regulation, health, and performance.
- Interpret HRV and biofeedback data using validated tools and metrics.
- Apply HRV biofeedback techniques in diverse contexts using real-time measurement tools.
- Evaluate the effects of different intervention components (e.g., resonance breathing, cognitive tasks, compassion exercises) on psychophysiological responses.
- Collaborate across disciplines to design, implement, and present evidence-based intervention plans and analyses.

Disclaimer: This course does not constitute professional certification in biofeedback or applied psychophysiology. While it introduces research-based concepts and hands-on practice, it does not provide the clinical hours or supervised experience required to apply biofeedback interventions in therapeutic or medical settings.

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3. Course Coordinator

Dr. Sigrún Þóra Sveinsdóttir, PhD
Psychophysiology & HRV Biofeedback

Dr. Sigrún Þóra Sveinsdóttir is a licensed psychologist, researcher, and educator specializing in psychophysiology, mental resilience, and heart rate variability (HRV) biofeedback. She holds a PhD in psychology, with doctoral research focusing on how autonomic regulation shapes cognitive performance, resilience, and functioning under stress. Her work bridges scientific research and applied practice, integrating HRV biofeedback into clinical, occupational, leadership, and performance-based settings, including work with police officers and individuals experiencing chronic stress, trauma, and long-term health conditions. She has held senior roles across healthcare, public health, and governmental systems, and actively publishes and presents international research within applied psychophysiology and biofeedback.

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4. Example of Readings and Resources

Core Texts:

- Lehrer, P., & Gevirtz, R. (2014). *Heart rate variability biofeedback: How and why does it work? Frontiers in Psychology*, 5, 756.
- Shaffer, F., & Ginsberg, J. (2017). *An overview of heart rate variability metrics and norms*. *Frontiers in Public Health*, 5, 258.
- Lehrer, P., Kaur, K., Sharma, A., Shah, K., Huseby, R., Bhavsar, J., & Zhang, Y. (2020). Heart rate variability biofeedback improves emotional and physical health and performance: A systematic review and meta analysis. *Applied psychophysiology and biofeedback*, *45*, 109-129.
- Steffen, P. R., & Moss, D. (2024). *Integrating Psychotherapy and Psychophysiology: Evidence-Based Assessment and Practice*. Springer. (Focus on Section III)
- Lehrer, P. (2024). The Importance of Including Psychophysiological Methods in Psychotherapy. *Applied Psychophysiology and Biofeedback*, 1-20

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5. Course Delivery and Study Schedule

The course is delivered as blended learning with structured team-based learning, daily individual practice, and an on-site intensive week.

Team structure and weekly coordination

Students are assigned to interdisciplinary teams prior to the first course date (21 May). Before the first live meeting, teams hold an initial meeting to agree on team roles (team lead and note-taker/secretary) and to set a weekly team meeting time for the duration of the course. Teams receive a team workbook (team handbook) and each student receives an individual workbook to support HRV practice and reflections.

Assessments and deliverables

- Group Assignment 1 (Team teaching, 27 May): Each team is assigned a topic/module to teach to the cohort in a 2-hour interactive session (requires active participation and use of a digital tool such as Mentimeter).
- On-site Team Experiment (1-5 June): Teams design and run a small within-subject experiment comparing HRV biofeedback alone versus HRV biofeedback combined with an additional component (e.g., self-compassion), including a challenge task and recovery.
- Group Assignment 2 (Capstone report + presentation, due during 6-17 June): Data analysis and applied mini-intervention rationale based on the on-site experiment.

- Individual deliverable: Daily HRV practice log and short reflective entries in the individual workbook across the course period.

Preliminary timeline (dates may include minor adjustments)

Week	Dates	Format	Focus / key activities
1	21-26 May	Online + team setup	Course orientation (intro video + live session). Overview of applied psychophysiology. Start HRV training.
2	27-31 May	Online (synchronous + asynchronous)	Team teaching. Continue HRV training. Online session (HRV training Week 2 + applied psychophysiology).
3	1-5 June	On-site intensive	HRV training Week 3 + applied labs. Team experiment design and data collection. Guest teaching: Erik Peper, John Hasslinger, Dr. Klumpers.
4	6-10 June	Online	Data analysis continued daily HRV practice. Team meetings. Draft capstone report.
5	11-17 June	Online	Team capstone presentations and submissions. HRV training Week 6 integration into daily life and reflection in the final session.

6. Attendance and Participation

- Active participation in both online and on-site components is mandatory
- Non-credit participants may attend lectures and practice labs but will not complete assessments or the capstone.

7. Academic Integrity

All submitted work must be original and adhere to APA 7th Edition citation standards. Plagiarism will be addressed per university policy.

Guest Faculty & Visiting Experts (Onsite week)

The course is developed and led by the course instructor, who is responsible for the overall pedagogical design, learning outcomes, assessment, and integration of theory, research, and practice throughout the course. During the on-site intensive week, selected guest experts contribute specialized teaching aligned with the course objectives.

Dr. Erik Peper - International leading figure in biofeedback and applied psychophysiology

Dr. Erik Peper is a leading figure in biofeedback and applied psychophysiology. With decades of experience as a researcher, educator, and clinician, he has helped shape modern biofeedback practice worldwide. He is a Professor at San Francisco State University, former President of the Association for Applied Psychophysiology and Biofeedback, and current President of the Biofeedback Federation of Europe.

Dr. Peper has authored numerous scientific articles and books on biofeedback, breathing, posture, stress regulation, and health optimization. His recent books include *TechStress: How Technology is Hijacking Our Lives* and the forthcoming *Cancer Reconsidered—Why Environment, Lifestyle, and Immunity Matter More Than We Thought*. He publishes the blog *The Peper Perspective* (www.peperperspective.com) and maintains a practice at BiofeedbackHealth (www.biofeedbackhealth.org). His teaching is internationally recognized for translating complex psychophysiological principles into practical, life-changing skills.

Contact information
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Dr. John Hasslinger – Neurofeedback & Self-Regulation Research

Dr. John Hasslinger is a researcher and educator specializing in neurofeedback and psychophysiological self-regulation. His work focuses on both experimental and applied investigations of neurofeedback protocols, brain regulation, and cognitive performance. He brings advanced expertise in neurofeedback methodologies and their integration within broader psychophysiological training frameworks, contributing to the development of evidence-based approaches for enhancing self-regulation in research, clinical, and applied performance settings.

Dr. Floris Klumpers – HRV Biofeedback, Virtual Reality & High-Stress Training

Dr. Floris Klumpers is a researcher working at the intersection of heart rate variability (HRV) biofeedback, immersive virtual reality, and stress–performance training. His work includes the development, implementation, and evaluation of VR-supported HRV biofeedback interventions designed for high-stress professional environments, including police training. His research emphasizes innovative, technology-enhanced applications of applied psychophysiology, with a strong focus on preparing individuals to perform effectively under pressure.